Failure of split-product approximations for time-dependent form sums

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Abstract

We give counterexamples to the stated exponential and resolvent split-product formulas for nonautonomous Schrodinger evolution. Narrow time spikes cause failure of the proposed limits while retaining the required common form domain and time regularity. A separate exponential counterexample uses uniformly bounded perturbations.

Repository target. This manuscript concerns AIM problem 917 at revision `5d7e26e8444e`. The precise scope is stated below.

1 Main result and proof

Outcome. The two assertions in Problem 917 are false under its stated hypotheses. The counterexample below has a bounded, finite-rank, self-adjoint perturbation at every individual time. The restricted forms are continuously differentiable in operator norm from the common form domain to its dual. Thus no convention concerning non-semibounded closed forms is involved. These assumptions allow time dependence that is smooth on the form scale while remaining badly behaved on fixed Hilbert-space vectors. The final strengthening even has uniformly bounded perturbations. The precise question is in the catalogue [1]; its source is [2, conditions (V), (V′), and Remark (3), equations (32)–(33)].

Theorem 1.1. *There exist a positive self-adjoint operator H_0 on a separable complex Hilbert space $\mathcal{H}$ and a family of bounded finite-rank self-adjoint operators $V(t)$, $0 \le t \le 1$, satisfying every hypothesis of Problem 917, for which the exponential split products at $(t,s) = (1,0)$ have no strong limit. The resolvent split products do not converge strongly, or weakly, to the specified unitary propagator $U(1,0)$.*

Proof. We first specify the sampling scales. Set

$$\alpha = \frac{\pi}{2}, \qquad \delta = \frac{1}{64}, \qquad \beta(s) = \begin{cases} (1-s^2)^2, & |s| \le 1, \\ 0, & |s| > 1. \end{cases}$$

The function β is C^1, satisfies $\beta(0) = 1$ and $0 \le \beta \le 1$, and has $|\beta'| \le 2$. Choose positive integers N_n recursively, beginning with $N_1 \ge 4$, with $N_n \ge 4N_{n-1}$. Once N_n is chosen, define

$$h_n = N_n^{-1}, \qquad \tau_n = 1 - h_n, \qquad g_n = \alpha N_n, \qquad w_n = \frac{2^{-n}}{1024 g_n}, \qquad v_n(t) = g_n \beta\Big(\frac{t-\tau_n}{w_n}\Big).$$

In choosing N_n, require also

$$\frac{1}{N_n}\sum_{j=0}^{N_n-1}\sum_{k<n} v_k(j/N_n) \le \delta. \tag{1}$$

This is a valid recursive choice: the finite sum $\sum_{k<n} v_k$ is already determined and continuous, and

$$\int_0^1 \sum_{k<n} v_k(t)\,dt \le \sum_{k<n} 2g_k w_k \le \frac{1}{512} < \delta.$$

Its left Riemann sums converge to this integral. Consequently arbitrarily large integers meet (1) and the growth requirement.

Write $I_n = (\tau_n - w_n, \tau_n + w_n)$. Since $w_n < h_n/10$ and $h_{n+1} \le h_n/4$, these intervals are disjoint. In fact all I_k with $k > n$ lie strictly to the right of τ_n. The uniform N_n-grid used in the question therefore samples I_n exactly once, at its last point $\tau_n = (N_n - 1)/N_n$, and samples none of the later intervals. The endpoint 1 is not sampled.

Let $\mathcal{H} = \ell^2(\mathbb{N}_0)$ with orthonormal basis $e_0, e_1, \ldots$. Define a diagonal positive self-adjoint operator by

$$H_0 e_0 = e_0, \qquad H_0 e_n = \lambda_n e_n, \qquad \lambda_n = \left(\frac{g_n}{w_n^2}\right)^2 \quad (n \ge 1).$$

Thus $\lambda_n \ge 1$, and its form domain and form are

$$Q = D(H_0^{1/2}), \qquad q_0[u] = |u_0|^2 + \sum_{n\ge 1} \lambda_n |u_n|^2.$$

Let $S_n = |e_0\rangle\langle e_n| + |e_n\rangle\langle e_0|$, zero on the orthogonal complement, and put

$$V(t) = \begin{cases} v_n(t) S_n, & t \in I_n, \\ 0, & t \notin \bigcup_n I_n. \end{cases}$$

In particular $V(1) = 0$. At every time, $V(t)$ is a bounded finite-rank self-adjoint operator. Its closed Hermitian form has domain all of $\mathcal{H}$.

We check the uniform form assumptions explicitly. For $u \in Q$ and $t \in I_n$,

$$\begin{aligned} |q_{V(t)}[u]| &\le \frac{g_n}{\sqrt{\lambda_n}} \beta\!\left(\frac{t-\tau_n}{w_n}\right)\left(|u_0|^2 + \lambda_n |u_n|^2\right) \\ &\le w_n^2 q_0[u]. \end{aligned} \tag{2}$$

Use $q_0^{1/2}$ as the norm on Q. Under the natural identifications of Q and its dual, the restricted form operator is represented by

$$B(t) = H_0^{-1/2} V(t) H_0^{-1/2} = w_n^2 \beta\!\left(\frac{t-\tau_n}{w_n}\right) S_n \quad (t \in I_n),$$

and by zero outside the intervals. Hence

$$\|B'(t)\| \le 2 w_n \qquad (t \in I_n). \tag{3}$$

Both β and β' vanish at the interval endpoints, so B is norm C^1 away from 1. At the accumulation point,

$$\frac{\|B(t)\|}{1-t} \le \frac{w_n^2}{h_n - w_n} \le \frac{2w_n^2}{h_n} \longrightarrow 0 \quad (t \in I_n,\ n \to \infty),$$

while (3) tends to zero. It follows that B is norm C^1 on $[0,1]$, with $B'(1) = 0$. Equivalently, $t \mapsto \widetilde{V}(t)$ is norm C^1 as a map into $\mathcal{L}(Q, Q^*)$. Every w_n is smaller than $1/8$, so (2)–(3) imply both of the catalogue's bounds with $a = 1/4$ and $b = 0$. Moreover

$$q_t[u] = q_0[u] + q_{V(t)}[u] \ge \frac{3}{4} q_0[u] \ge \frac{3}{4}\|u\|^2.$$

The standing positivity condition holds with $c = 3/4$. The hypotheses therefore provide the unitary form propagator $U(t, s)$.

We next examine the exponential product on the grid of size N_n. Denote by K_n^E the product of its first $N_n - 1$ factors, omitting the last factor at τ_n. All perturbations sampled in K_n^E come from the earlier spikes $k < n$. Telescoping the unitary factors and using $\|e^{-ihV} - I\| \le h\|V\|$ gives

$$\|K_n^E - e^{-i\tau_n H_0}\| \le h_n \sum_{j=0}^{N_n-2} \|V(jh_n)\| \le \delta. \tag{4}$$

At the last grid point, $V(\tau_n) = g_n S_n$ and $h_n g_n = \pi/2$, whence

$$e^{-ih_n V(\tau_n)} e_0 = -ie_n.$$

Since $H_0 e_0 = e_0$, (4) implies

$$P_{N_n}^E e_0 = c_n e_n + r_n, \qquad c_n = -ie^{-i\tau_n} e^{-ih_n\lambda_n}, \qquad |c_n| = 1, \quad \|r_n\| \le \delta.$$

Consequently, whenever $m \ne n$,

$$\|P_{N_n}^E e_0 - P_{N_m}^E e_0\| \ge \sqrt{2} - 2\delta > 0.$$

The exponential products are not strongly Cauchy, and therefore have no strong limit.

Finally, let $R_n = (I + ih_n H_0)^{-1}$ and let K_n^R be the analogous resolvent prefix with the last factor omitted. All factors are contractions, and $\|(I + ihV)^{-1} - I\| \le h\|V\|$. The same telescoping argument yields

$$\|K_n^R - R_n^{N_n-1}\| \le \delta.$$

Thus $K_n^R e_0 = b_n e_0 + s_n$ with $|b_n| \le 1$ and $\|s_n\| \le \delta$. On the two-dimensional span of e_0, e_n,

$$(I + i\alpha S_n)^{-1} e_0 = \frac{e_0 - i\alpha e_n}{1 + \alpha^2}, \qquad \|(I + i\alpha S_n)^{-1} e_0\| = (1 + \alpha^2)^{-1/2}.$$

The final R_n is a contraction, so

$$\|P_{N_n}^R e_0\| \le (1 + \alpha^2)^{-1/2} + \delta < 1.$$

As $U(1, 0)$ is unitary,

$$\|P_{N_n}^R e_0 - U(1, 0)e_0\| \ge 1 - (1 + \alpha^2)^{-1/2} - \delta > 0.$$

This excludes the proposed strong limit. It excludes weak convergence to $U(1, 0)$ as well: for contractions P_n and a unitary U, weak convergence $P_n \to U$ implies strong convergence, because

$$\|P_n x - Ux\|^2 \le 2\|x\|^2 - 2\,\mathrm{Re}\langle P_n x, Ux\rangle \longrightarrow 0.$$

□

An independent construction of the propagator. In this example the existence of a unitary propagator can also be seen without an abstract form-evolution theorem. Indeed,

$$\int_0^1 \|V(t)\|\,dt = \left(\int_{-1}^1 (1 - s^2)^2\,ds\right) \sum_{n\ge1} g_n w_n = \frac{16}{15}\frac{1}{1024} = \frac{1}{960}.$$

The operator-valued function is Bochner measurable: on each spike it is a continuous scalar multiple of a fixed finite-rank operator, and its range lies in a separable subspace of $\mathcal{B}(\mathcal{H})$. The interaction-picture perturbation $W(t) = e^{itH_0} V(t) e^{-itH_0}$ is also Bochner measurable and has

the same integrable norm. On I_n its two matrix entries merely acquire the continuous phases $e^{\pm it(\lambda_n-1)}$. The norm-convergent Dyson series therefore solves

$$Z(t,s) = I - i\int_s^t W(\tau)Z(\tau,s)\,d\tau.$$

It is unitary because $W(t)$ is self-adjoint almost everywhere. Hence $e^{-itH_0}Z(t,s)e^{isH_0}$ is a unitary propagator. It agrees with the form propagator: testing the form equation against finite-support vectors transported by e^{-itH_0} cancels the free generator and gives the same integral equation; its solution is unique. By density this identifies the propagators on all of $\mathcal{H}$. Unitarity in the integral equation gives the stronger quantitative statement

$$\|U(1,0) - e^{-iH_0}\| = \|Z(1,0) - I\| \le \frac{1}{960}.$$

Thus the actual evolution is very close to the free evolution even though the selected grids produce the failures proved above.

The obstruction is an exact sampling effect. Each narrow time spike is small, together with its first derivative, on the form scale, but its peak acts strongly on a fixed low-energy vector in the Hilbert space. A selected uniform grid hits that peak and transfers the vector to a new high-energy basis vector. Earlier sampled spikes have uniformly small total effect by construction. The example even has a common operator domain $D(H_0)$ for all $H_0 + V(t)$, since each $V(t)$ is bounded. The form hypotheses do not control the action of the sampled perturbations on fixed Hilbert-space vectors. The following strengthening shows that a uniform operator bound alone still does not ensure the exponential limit.

1.1 The exponential product can fail even for uniformly bounded perturbations

The exponential obstruction has a stronger version. A uniform bound on $\|V(t)\|$ alone does not repair the statement; time regularity in the Hilbert-space operator topology matters as well.

Theorem 1.2. *For every $\varepsilon > 0$, there is a family satisfying all hypotheses of Problem 917, with every $V(t)$ self-adjoint and finite rank, such that*

$$\sup_{0\le t\le 1}\|V(t)\| \le 1, \qquad \int_0^1 \|V(t)\|\,dt \le \varepsilon,$$

but its exponential split products at $(t,s) = (1,0)$ have no strong limit. The restricted forms are continuously differentiable in operator norm, and $V(t)$ is continuous in the weak operator topology.

Proof. Fix $\delta = 1/100$ and put $\varepsilon_0 = \min\{\varepsilon, \delta/8\}$. Let $\beta(s) = (1-s^2)^2$ on $[-1,1]$ and zero elsewhere, as before. We recursively choose integers N_n, finite sets $J_n \subset \{0,\dots,N_n-1\}$, and widths $w_n > 0$.

Suppose the earlier groups have been chosen. Let E_{n-1} be the union of their closed bump intervals, and let

$$Z_{n-1} = \{0,1\} \cup \bigcup_{k<n}\{j/N_k : 0 \le j \le N_k - 1\}.$$

The set E_{n-1} is a finite union of intervals; its length will be at most ε_0. Its indicator is Riemann integrable. Choose N_n sufficiently large that

$$\frac{1}{N_n}\#\{0 \le j < N_n : j/N_n \in E_{n-1}\} \le \frac{\delta}{4}, \qquad \frac{\#Z_{n-1}}{N_n} \le \frac{\delta}{4}.$$

This is possible because $|E_{n-1}| \le \varepsilon_0 \le \delta/8$ and Z_{n-1} is finite. Require also $N_n > N_{n-1}$ when $n > 1$, and take $1/N_n$ smaller than every positive point of $E_{n-1} \cup Z_{n-1}$. The latter requirement is possible because this finite union of closed intervals and points stays a positive distance from zero after zero itself is removed. Set

$$J_n = \{0 \le j < N_n : j/N_n \notin E_{n-1} \cup Z_{n-1}\}, \qquad m_n = \#J_n.$$

Then $m_n/N_n \ge 1 - \delta/2$, and $1 \in J_n$. Choose $w_n < 1/(4N_n)$ so small that the closed intervals

$$[j/N_n - w_n, j/N_n + w_n], \qquad j \in J_n,$$

are pairwise disjoint, lie in $(0,1)$, and avoid $E_{n-1} \cup Z_{n-1}$. Also impose

$$2N_n w_n \le \varepsilon_0\, 2^{-n}.$$

All these requirements are possible because there are finitely many chosen centers, each at positive distance from the closed forbidden set. They ensure that all bump intervals in all groups are disjoint and that

$$|E_{n-1}| \le \sum_{k<n} 2m_k w_k \le \varepsilon_0,$$

justifying the induction. Define

$$b_n(t) = \sum_{j \in J_n} \beta\Big(\frac{t - j/N_n}{w_n}\Big).$$

Each b_n is C^1, with $0 \le b_n \le 1$ and $\|b_n'\|_\infty \le 2/w_n$. Every later group vanishes at every point of the N_n-grid, by its avoidance of Z_{k-1}.

On $\ell^2(\mathbb{N}_0)$ let $H_0 e_0 = e_0$ and $H_0 e_n = \lambda_n e_n$, choosing

$$\lambda_n = 1 + 2\pi k_n N_n, \qquad k_n \in \mathbb{N},$$

so large that

$$\lambda_n^{-1/2}(1 + 2/w_n) \le 2^{-n}/16.$$

Let $S_n = |e_0\rangle\langle e_n| + |e_n\rangle\langle e_0|$, and put

$$V(t) = \sum_{n \ge 1} b_n(t) S_n.$$

At each time at most one summand is nonzero. Thus $V(t)$ is bounded, self-adjoint and of rank at most two, with $\|V(t)\| \le 1$. Moreover,

$$\int_0^1 \|V(t)\|\, dt \le \sum_{n \ge 1} 2m_n w_n \le \varepsilon_0 \le \varepsilon.$$

For $Q = D(H_0^{1/2})$, the restricted form is represented by

$$B(t) = H_0^{-1/2} V(t) H_0^{-1/2} = \sum_{n \ge 1} \lambda_n^{-1/2} b_n(t) S_n.$$

Both this series and its derivative series converge uniformly in operator norm, and $\sup_t \|B(t)\| \le 1/16$, $\sup_t \|B'(t)\| \le 1/16$. Hence B is norm C^1. The form bounds hold, for example, with $a = 1/8$ and $b = 0$, while $q_0 + q_{V(t)} \ge (15/16) q_0 \ge (15/16)\|\cdot\|^2$. This verifies every form hypothesis. Matrix elements are continuous first for vectors in Q, by form continuity, and then

for arbitrary Hilbert-space vectors by density and the uniform bound on $\|V(t)\|$. Thus V is weak-operator continuous. It is not strongly continuous at zero: $V(0)e_0 = 0$, while $V(1/N_n)e_0 = e_n$ and $1/N_n \to 0$. Its Bochner measurability and the existence of the unitary propagator follow as in the preceding Dyson construction.

Consider the product on the N_n-grid, with $h_n = 1/N_n$. At a node $j \in J_n$, its perturbation is exactly S_n. At all other nodes it is an earlier perturbation or zero; no later group contributes. Compare the product with the ideal one that keeps S_n at the nodes of J_n and uses zero at all other nodes. Telescoping products of unitaries and using $\|e^{-ihV} - I\| \le h\|V\|$ bounds their operator-norm difference by

$$\frac{1}{N_n}\#\{0 \le j < N_n : j/N_n \in E_{n-1}\} \le \frac{\delta}{4} \le \delta.$$

On $\operatorname{span}\{e_0, e_n\}$, the free step is precisely $e^{-ih_n}I$, since $h_n(\lambda_n - 1) = 2\pi k_n$. It therefore commutes with each retained S_n-step on this subspace. Writing $a_n = m_n/N_n \in [1-\delta, 1]$, the ideal product sends e_0 to

$$v_n = e^{-i}\big(\cos(a_n)e_0 - i\sin(a_n)e_n\big).$$

Consequently $\|P^E_{N_n}e_0 - v_n\| \le \delta$. For $m \ne n$,

$$\|P^E_{N_n}e_0 - P^E_{N_m}e_0\| \ge \sqrt{2}\sin(1-\delta) - 2\delta > 0.$$

The sequence is not strongly Cauchy. Finally, the Dyson estimate gives $\|U(1,0) - e^{-iH_0}\| \le \varepsilon_0$, so the actual evolution can be made arbitrarily close to the free evolution while the splitting still has this fixed nonconvergence gap. □

References

[1] MColbrook/AIM, *AIM 917: Split products for time-dependent form-sum Schrödinger evolution*, repository statement at revision `5d7e26e8444e`. https://github.com/MColbrook/AIM/blob/5d7e26e8444eb98de91988454564e45e011107c4/problems/917-split-products-for-time-dependent-form-sum-schrodinger-evolution.md.

[2] P.-A. Vuillermot and W. F. Wreszinski, *Product approximations for solutions to a class of evolution equations in Hilbert space*, Portugaliae Math. **68** (2011), 317–343. doi:10.4171/PM/1894.